API Contract Specification

Hasnahena Platform

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Author:	Md Saif Ahmed

RESTful API specifications
for frontend and service integration

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Document Control

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Ver.	Date	Author	Changes
1.0	March 22, 2026	Md Saif Ahmed	Initial release

1 API Conventions

1.1 Base Configuration

Property	Value
Base URL	`https://api.hasnahena.org/v1`
API Version	v1
Content-Type	`application/json`
Character Encoding	UTF-8

1.2 Authentication

All protected endpoints require a Bearer token in the Authorization header:

```
Authorization: Bearer <JWT_TOKEN>
```

Token type: JWT (JSON Web Token) compatible with session-based gateway policy.

1.3 Response Envelope

All API responses follow a consistent envelope structure:

```
{
    "success": true | false,
    "data": { ... },
    "error": {
        "code": "ERROR_CODE",
        "message": "Human readable message"
    },
    "meta": {
        "timestamp": "ISO-8601",
        "requestId": "uuid"
    }
}
```

2 Authentication Endpoints

Endpoint	Method	Description
`/auth/login`	POST	Authenticate via phone/email and password
`/auth/change-password`	POST	Mandatory first-login password update
`/auth/logout`	POST	Invalidate session/token
`/auth/refresh`	POST	Refresh access token

3 Member and Registration Endpoints

Endpoint	Method	Description
`/registrations`	POST	Submit registration request

Endpoint	Method	Description
/admin/registrations/{id}/approve	POST	Approve registration (Admin)
/admin/registrations/{id}/reject	POST	Reject registration (Admin)
/members/me	GET	Retrieve member profile and summary
/members/me/nominee	PUT	Update nominee details

4 Transaction and Loan Endpoints

4.1 Deposit & Withdrawal

Endpoint	Method	Description
/transactions/deposit-requests	POST	Member creates deposit request
/transactions/withdrawal-requests	POST	Member creates withdrawal request
/worker/transactions/{id}/process	POST	Worker marks request processed
/member/transactions/{id}/confirm	POST	Member confirms transaction
/member/transactions/{id}/dispute	POST	Member raises dispute

4.2 Loan Management

Endpoint	Method	Description
/loans/applications	POST	Submit loan application
/loans/{id}/guarantor-approval	POST	Guarantor accepts/rejects
/admin/loans/{id}/decision	POST	Admin approves/rejects loan
/loans/{id}/disburse	POST	Worker records disbursal
/loans/{id}/installments	POST	Record installment payment
/admin/loans/{id}/settle-default	POST	Execute settlement waterfall

5 Donation and Reporting Endpoints

Endpoint	Method	Description
/donations	POST	Create donation by fund type
/dashboard/member	GET	Member dashboard aggregates
/dashboard/worker	GET	Worker pending queue and SLA metrics
/dashboard/admin	GET	Admin operational and risk indicators

Endpoint	Method	Description
/audit/logs	GET	Append-only audit query (Admin/Auditor)

6 Error Model

6.1 HTTP Status Codes

Code	Status	Description
200	OK	Request successful
201	Created	Resource created successfully
400	Bad Request	Validation failure (malformed request)
401	Unauthorized	Authentication required or failed
403	Forbidden	Insufficient role permissions
404	Not Found	Entity/resource not found
409	Conflict	State conflict (e.g., already confirmed/disputed)
422	Unprocessable Entity	Business rule violation (e.g., eligibility cap exceeded)
500	Server Error	Internal server error

6.2 Error Response Example

```
{
    "success": false,
    "data": null,
    "error": {
        "code": "ELIGIBILITY_CAP_EXCEEDED",
        "message": "Requested loan amount exceeds your eligibility",
        "details": {
            "requested": 100000,
            "maxAllowed": 50000
        }
    },
    "meta": {
        "timestamp": "2024-01-15T10:30:00Z",
        "requestId": "abc123-def456"
    }
}
```

7 Appendix: Common Error Codes

Error Code	Description
VALIDATION_ERROR	Request body validation failed
AUTH_INVALID_CREDENTIALS	Incorrect email/phone or password
AUTH_TOKEN_EXPIRED	JWT token has expired
MEMBER_NOT_FOUND	Member ID does not exist
LOAN_ELIGIBILITY_FAILED	Member does not meet loan criteria
WITHDRAWAL_BLOCKED	Active loan or guarantee prevents withdrawal
DISPUTE_ALREADY_RAISED	Transaction already has an active dispute
GUARANTOR_DECLINED	Guarantor rejected the guarantee request
INSUFFICIENT_FUNDS	Pool liquidity insufficient for operation